

Alessandro Alfano

Software Engineer · Systems Integration



Integration between systems, and the distributed services around them. Diverse background. Interested in open source, and in who the software ends up serving.

Contact Information

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Professional Experience

Application Engineer

Microtecna Engineering | Novazzano, Switzerland *2024-04 - Present*

Swiss engineering company specializing in high-tech industrial solutions, robotics, and automated systems

Responsibilities:

- Design and development of software applications and integration layers for automated systems, including architecture definition, communication protocols, and data models
- Contribution to the evolution of the company's application framework (.NET/WPF), focusing on modular architecture, extensibility, and web technology integration
- R&D activities: evaluation and adoption of new data storage technologies, CI/CD and containerization practices

- Management and evolution of company's private git infrastructure, introduction to private Package/Container Registry and advanced Git automatic workflows
- Internal reference for Linux environments, web technologies, and AI-assisted development workflows

Technologies: Git, GitLab CI/CD, Docker, InfluxDB, Linux, gRPC, MQTT, Redis, .NET, WPF

Focus: Reference point for application software, R&D, and development infrastructure

Software Architect

YouPower AG | Mezzovico, Switzerland *2023-10 - 2024-04*

Swiss energy technology company focused on smart metering and energy community solutions

Responsibilities:

- Led the complete redesign and rewrite of a distributed system for electricity consumption/production data collection and processing
- Coordinated development team and defined end-to-end system architecture
- Designed meter acquisition layer using Modbus protocol and IoT router configuration
- Architected on-premise cloud infrastructure with VMs and Docker Swarm
- Developed data visualization platform using Angular and NestJS
- Implemented automated billing data generation and OCPP server for EV charging stations

Technologies: Angular, NestJS, TypeScript, Docker Swarm, MongoDB, Redis, Modbus, RouterOS, WireGuard, OCPP

Focus: Architecture redesign of the data platform, from meter acquisition to billing

Full Stack Developer

Forge Srl | Varese, Italy *2022-02 - 2023-09*

Italian software house specializing in cloud-native solutions and enterprise integrations

Responsibilities:

- Contributed to Leapp, an open-source application for secure cloud access management in DevOps environments
- Designed and implemented AWS-based solutions using TypeScript, including e-commerce-to-ERP integration flows
- Developed microservices-oriented document processing system (Spring Boot) for Italian electronic invoicing standards

Technologies: TypeScript, AWS, Spring Boot, Java, Node.js

Focus: Open-source work on Leapp and cloud integrations for enterprise clients

Software Developer

Arsinform Srl | Milan, Italy *2021-05 - 2022-01*

Italian company specializing in industrial automation and warehouse management software

Responsibilities:

- Contributed to design and evolution of automation systems for automated warehouses
- Integrated and optimized mission management system for a leading Italian food producer/distributor
- Conducted operational research for technology stack modernization

Technologies: Oracle, PL/SQL

Focus: Mission management system for automated warehouses

Education

B.Sc. Computer Engineering

Politecnico di Milano - Milan, Italy *2020 - 2026*

Computer engineering, software development, system architecture

Medicine and Surgery (partial)

University of Milan - Milan, Italy *2014 - 2020*

Medicine and surgery

Classical Studies Diploma (Maturità Classica)

Liceo Classico Alessandro Volta - Como, Italy *2009 - 2014*

Latin, Ancient Greek, classical literature, philosophy

Core Competencies

Software Architecture & System Design

Design of modular systems: boundaries, responsibilities, data models, and how a system is meant to change over time.

Key Skills: System Design, Modular Architecture, Technical Strategy, Domain Modeling

Complex System Integration

Integration of systems across automation, IoT, cloud services, and enterprise platforms: communication protocols, data flows, and orchestration patterns.

Key Skills: System Integration, Protocol Design, Data Orchestration, Heterogeneous Systems

Distributed Systems & Cloud-Native Architectures

Distributed systems, microservices, containerized environments, and hybrid cloud/on-prem infrastructures.

Key Skills: Microservices, Containerization, Cloud-Native, Hybrid Infrastructure

DevOps & Engineering Practices

Git workflows, CI/CD pipelines, containerization with Docker, and development infrastructure management.

Key Skills: CI/CD, GitOps, Containerization, Infrastructure as Code

Applied Research & Technology Evaluation

Evaluation and adoption of new technologies, integration patterns, and AI-assisted workflows.

Key Skills: R&D, Technology Evaluation, Innovation, Continuous Improvement

Technical Skills

Architecture & Design

System Architecture, Microservices Design, Domain-Driven Design, Event-Driven Architecture, API Design

Programming Languages

- **Primary:** TypeScript, JavaScript, Java, C#, Lua
- **Secondary:** Python, Rust, SQL
- **Scripting:** Bash, PowerShell

Frameworks & Libraries

- **Backend:** .NET, NestJS, Node.js, Spring Boot
- **Frontend:** Angular, Blazor, Vue

Databases

- **Relational:** PostgreSQL, MySQL, SQLite
- **NoSQL:** MongoDB, Redis, InfluxDB
- **Cloud:** DynamoDB

Infrastructure & DevOps

- **Containerization:** Docker, Docker Swarm, Docker Compose, Kubernetes
- **CI/CD:** GitLab CI/CD, GitHub Actions
- **Cloud Platforms:** AWS

Integration & Protocols

gRPC, MQTT, REST, WebSocket, OPC UA, Modbus TCP, Signal R

Courses

- **AWS Solutions Architect – Associate (Training Course)** - beSharp Srl (2022-11)
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Projects

Main Projects

Alphamon

Game Framework - In progress · repository currently private

Collectible monster-fighting RPG starter kit in Lua/Love2D, with dependency injection, TCP multiplayer, a battle system, and dialog management. MIT-licensed, to be released as open source.

Features:

- Dependency Injection
- Multiplayer Networking
- Battle System
- Dialog Management

Technologies: Lua, Love2D

Multi-Protocol Integration Framework

Integration Framework - Professional project (implementation proprietary)

Framework/application template for multi-protocol integration across distributed, heterogeneous environments. Supports OPC UA, Redis, REST, gRPC, and other protocols for heterogeneous system communication.

Technologies: .NET, OPC UA, Redis, gRPC, REST

AML (Alessandro's Music Language)

Creative Tool - Public repository · MIT

Text-based music notation system for chiptune composition, with a Game Boy Color-style synthesis framework built on ChuckK and Python CLI utilities.

Technologies: ChuckK, Custom DSL

University Projects

GathorApp

Web Application

Outing and event management system with geolocation, JWT authentication, real-time chat, and vouchers.

Technologies: Spring Boot, Angular, JWT, WebSocket

<https://github.com/lysandre995/gathorapp>

Green Wheels

Web Application

Car pooling application, split into frontend, backend, and shared-core repositories.

Technologies: TypeScript, Angular, Node.js

<https://github.com/lysandre995/green-wheels>

Other Public Projects

Lorenzo Orsenigo Website

Website

Website for percussionist Lorenzo Orsenigo.

Technologies: Nunjucks, HTML, CSS, JavaScript

<https://github.com/lysandre995/lorenzo-orsenigo-website>

Personal Portfolio

Website

Personal website and CV platform with JSON-based data backend.

Technologies: TypeScript

<https://alessandroalfano.com>

Waterfall Enterprise

Website

Site of the fictional company that appears in the theatre production Waterfall, written and directed by Ilaria Casati.

Technologies: Eleventy, Nunjucks, Bootstrap, JavaScript

<https://waterfall-enterprise.org>

Open Source Contributions

Leapp

Noovolari

Secure cloud access management for DevOps environments (AWS, Azure, GCP)

Feature development during the Forge period (2022-2023)

<https://github.com/Noovolari/leapp>

Languages

- **Italian:** Native
 - **English:** C1 (Professional)
 - **French:** A2 (Elementary)
 - **Latin:** Reading
 - **Ancient Greek:** Reading
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Interests

Creative Pursuits

- **Music:** Singing and playing in a rock band; chiptune composition; music notation systems
- **Game Development:** Indie game development in Lua/Love2D

Physical Activities

- **Sailing**
 - **Soccer Coaching:** Youth team coaching
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Theatre

Communication with Teatro Sociale di Como

Compagnia Teatrale Trebisonda | 2026 - Present

Contact person for the company's relationship with Teatro Sociale di Como. Current work covers workshops and "Theatron - una sfida a teatro", part of the Stravolta campaign of the 2026/2027 season.

Waterfall, by Ilaria Casati

Compagnia Teatrale Trebisonda | 2025 - 2026

Production staged in four performances between 2025 and 2026. The site of the fictional company featured in the play is listed under Projects.

<https://waterfall-enterprise.org>

Workshop activities

Mumble Teatro | 2021 - 2025

Workshop cycles with the association, each ending in a final production; four in total.

Values

- Technology for social good
 - Open source and transparency
 - Environmental sustainability
 - Community empowerment
 - Ethical engineering practices
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I authorize the processing of personal data in accordance with Legislative Decree 101/2018 and GDPR (EU Regulation 2016/679)

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